

Name _____

Year 12 Human Biology General

Task

Bacteria and Fungi Practical

This assessment consists of two parts.

Part a – Processing second hand data.

Part b – Planning an investigation.

Part a Surface Testing for Pathogens

(10 marks)

The students were given four petri dishes, each containing agar. They were asked to select four areas of the school where they would expect to find either bacteria or fungi.

They then used cotton swabs to take sample from the surfaces and brushed the swabs across different petri dishes. One area per dish. The petri dishes were sealed and incubated in an oven for several days. They also set up a fifth dish as a control treatment.

Two weeks later the petri dishes were examined. The petri dishes were NOT opened. The students identified any colony as either bacteria or fungi and then recorded the number of colonies present in a table.

Bacteria colonies appear waxy and the may have a clear ring around the colony as bacteria can produce a toxin that kills competing pathogens.

Fungal colonies appear furry. Different species may be different colours.

After the observations had been made the unopened petri dishes were autoclaved (boiled in a pressure cooker) and then burnt.

Table of Bacterial and Fungal Colonies on Surfaces in the School

Surface Location	Number of Bacteria	Number of Fungi
Gym showers	10	1
Toilet bowl	0	0
Toilet wall mirror	22	15
Classroom desk	0	0
Control	0	0

1. Graph the data shown in this table in the most appropriate form. (5 marks)

2. Why were the petri dishes sealed and students not allowed to open the petri dishes to get a clearer view of the colonies? (1 mark)

3. Why do you think the students chose to test the shower in the gym? (1 mark)

4. Some girls had been observed putting on lip balm and then kissing their reflection in the toilet mirror. How might this observation affect the results? (1 mark)

5. Suggest reasons to explain why the results differed from surface to surface? (2 marks)

Part b The Effectiveness of Handwashing with Water and Hand Sanitiser.

(10 marks)

Background Information

Washing hands is a recommended way of reducing the spread of bacteria, particularly when handling food or sick people. The use of hand sanitisers is increasing in popularity. Is the use of these hand sanitisers more effective than washing hands?

This activity is to test the effect of washing hands and using hand sanitisers on the bacterial population of the skin. The presence of bacteria on the skin can be demonstrated by pressing the fingers into the surface of the agar in a sterile petri dish. The agar in the petri dish contains nutrients for any micro-organisms that might grow on its surface.

After pressing the fingers on the agar the plates should be sealed and then incubated for a number of days in a warm space. Any bacteria transferred from the fingers will grow into colonies that can be seen by the human eye. The more colonies growing the more bacteria that were on the skin.

You are to design and plan to test the effectiveness of hand sanitisers and hand washing. Then you should answer the following questions.

1. What is the Independent Variable for this test? (1 mark)

2. What is the Dependent Variable for this test? (1 mark)

3. What hypothesis is being tested? (1 mark)

4. Name two variables that will need to be controlled to make it a valid test? (2 marks)

5. How will you distinguish between

(2 marks)

A bacterial and a Fungal colony? _____

Different species of bacteria or fungi? _____

A previous group of students carried out this investigation. Here are their results.

Day number	Agar Plate/Hand Washing	Appearance
1	Nothing	Misty spots
1	Sanitiser	Small waxy spots
1	Water	Misty
3	Nothing	Furry strings. Black dots
3	Sanitiser	Small waxy spots
3	Water	Large waxy spots Small yellow dots
5	Nothing	Black furry Waxy dots
5	Sanitiser	Furry white blob Small waxy spots
5	Water	Furry white blob Waxy spots

6. Which group acted as the Control Treatment for the investigation?

(1 mark)

7. What conclusion(s) can be drawn from these results?

(2 marks)

Cite evidence for your answer.
